

Second Cast: Functionally Graded Composite Walls from Recycled Concrete and Foam-Crete

Abstract

Turning recycled concrete rubble into lightweight, modular, and deployable building units not only adds value to waste, but is also a more sustainable approach to keeping building materials in use and circulation. Concrete remains indispensable to the built environment due to its high compressive strength, yet it constitutes the largest component of construction and demolition debris (U.S. EPA 2018, 16). Conventional demolition practices typically downcycle concrete into low-grade aggregate for use as road base or backfill, while substantial quantities are ultimately sent to landfills. This project addresses our recycling habits, and proposes an integrated, high-performance composite wall system that offers measurable embodied carbon reductions.

Keywords: *Recycled Concrete, Rubble, Foam-crete, Composite Wall, Lightweight*

1. Introduction and Background

Second Cast proposes a material strategy intended to disrupt downcycling. Greater value is argued to exist with resources already extracted (see Figure 1). Within the building industry, new systems of collection, sorting, reuse, and processing therefore can be developed to extend the life and utility of existing material stock (Heisel, Hebel, and Webster 2022). Building upon this premise, *Second Cast* is a material reuse strategy leveraging concrete densities, coalesced into a new way of manufacturing wall panels. This project builds on P. Bedarf's work on mineral foam, combining recycled concrete rubble with lighter foam-concrete (foam-crete) to create high-performance pre-cast concrete panels (Bedarf et al. 2023). Similar to the benefits of Autoclave Aerated Concrete (AAC) masonry block, foam mechanically mixed with concrete integrates acoustic, thermal, and fire-resistance properties (Khunt et al. 2023).

2. Foam-Concrete Mixture Testing

Central to the foam-crete mix is precise control of density. The foam is produced by aerating a foaming solution (Drexel FM-160) with water under compression and passing it through steel wool to create dense, stable foam (see Figure 2). To preserve its aeration, the foam is manually folded into the concrete rather than a mixer. An accelerator (Daraset 400) is introduced to control setting time, allowing the foam to be incorporated approximately 15 - 20 minutes after the base concrete is fully mixed. To prevent the foam from collapsing during integration, the mixture must reach a sufficient stiffness. The addition of foam increases the concrete's volume by approximately 85% without significantly increasing its weight, thereby reducing the overall density of the matrix (see Figure 3).

3. Computationally Optimized Wall Panel

The wall design explores a 4'-0" x 4'-0" x 0'-4" panel with an offset rectangular opening, creating a non-uniform element to study a compression-driven form-finding optimization (see Figure 4,5,6). The Peregrine plugin for Grasshopper was used to iteratively model loading forces and redistribute material in response to those forces. By simulating both vertical dead loads and lateral wind loads, the system iteratively minimizes volume while preserving load paths, resulting in a materially efficient compression network. The final geometry demonstrates how topology optimization can rethink a monolithic, homogeneous panel with a differentiated structural logic, placing load-bearing concrete where needed and replacing non-structural filler zones with foam-crete, while maintaining stability and providing additional thermal, acoustic, and fire-resistant benefits (see Figure 7,8).

4. Environmental Impact and Application

The study evaluates the embodied carbon implications of the optimized composite wall panels under a defined set of manufacturing and transportation assumptions explained in Figure 9. The analysis found that the final panel design achieves up to a 33% reduction in mass and approximately 30% reduction in embodied carbon relative to a baseline solid concrete panel with no topological or material optimization, considering only manufacturing and transportation impacts. The scope of this analysis centers on the reuse of recycled concrete rubble as aggregate and the strategic use of foam-crete to reduce overall weight without compromising structural capacity (See Figure 10). Future applications may extend this approach to precast floor slabs, column systems, and other circular construction strategies, including design for assembly and disassembly.

References

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Appendix

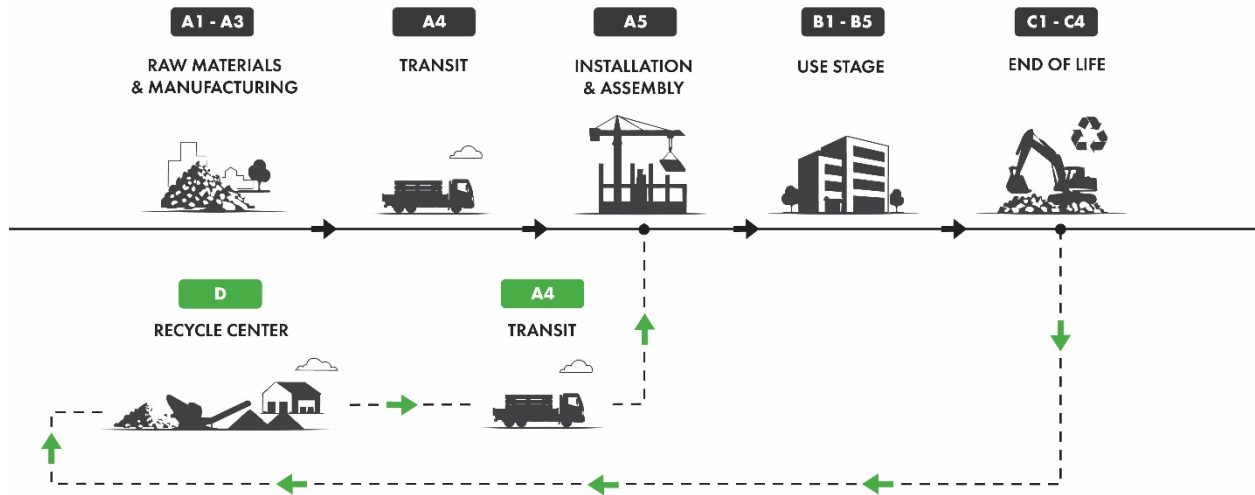


Figure 1. Concrete Proposed Life Cycle. Among the many adjustable parameters, the primary design objective is to reduce concrete use without sacrificing performance, while incorporating recycled concrete rubble as aggregate and accounting for off-site manufacturing considerations, including transportation and deployment.



Figure 2. Foam Generator. Foaming agent compressed and passed through a dense amount of steel wool creating small, structured foam. Mechanically mixed with concrete once the mix reaches a certain consistency.

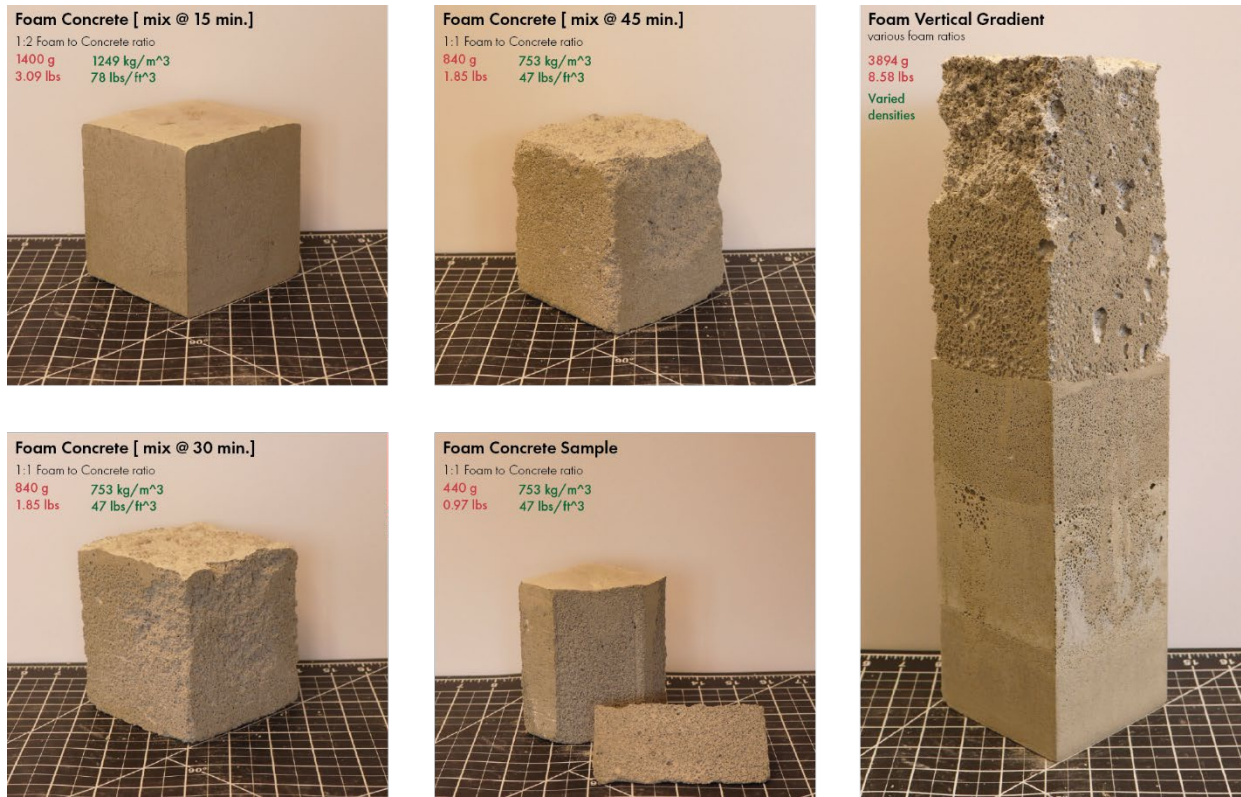


Figure 3. Mixture Testing. Twenty tests were developed to control different densities of the foam concrete, including density grading across a single cast to test bonding adhesion between the different mixes. Concrete mixture variables including accelerator volume, sand, cement, polymer, and timing were all tested. Including the mixture of concrete and foam at 15-minute intervals after the addition of accelerator to the concrete.

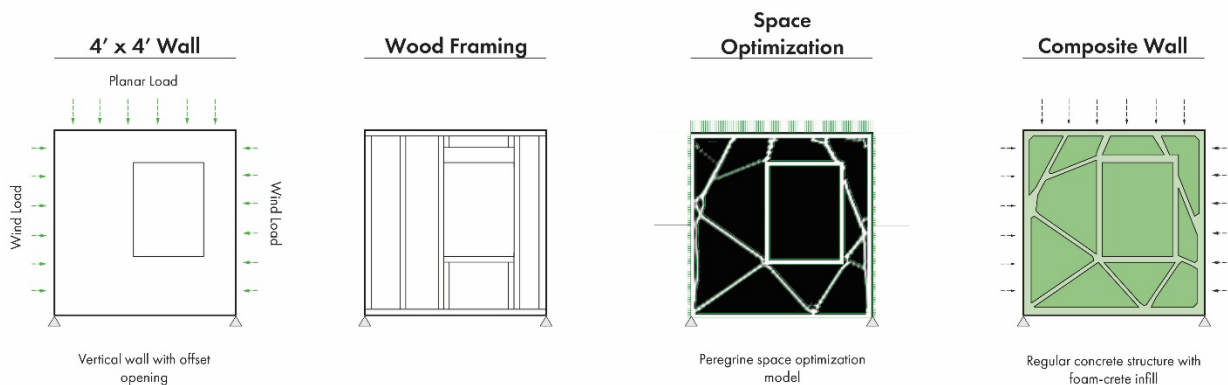


Figure 4. Title. Using Peregrine plugin through Grasshopper and other software, the wall is simulated with vertical dead loads and lateral wind loads. Adjusting tree branch thickness and the ratio of concrete to foam-crete, the result is a materially efficient compression network.

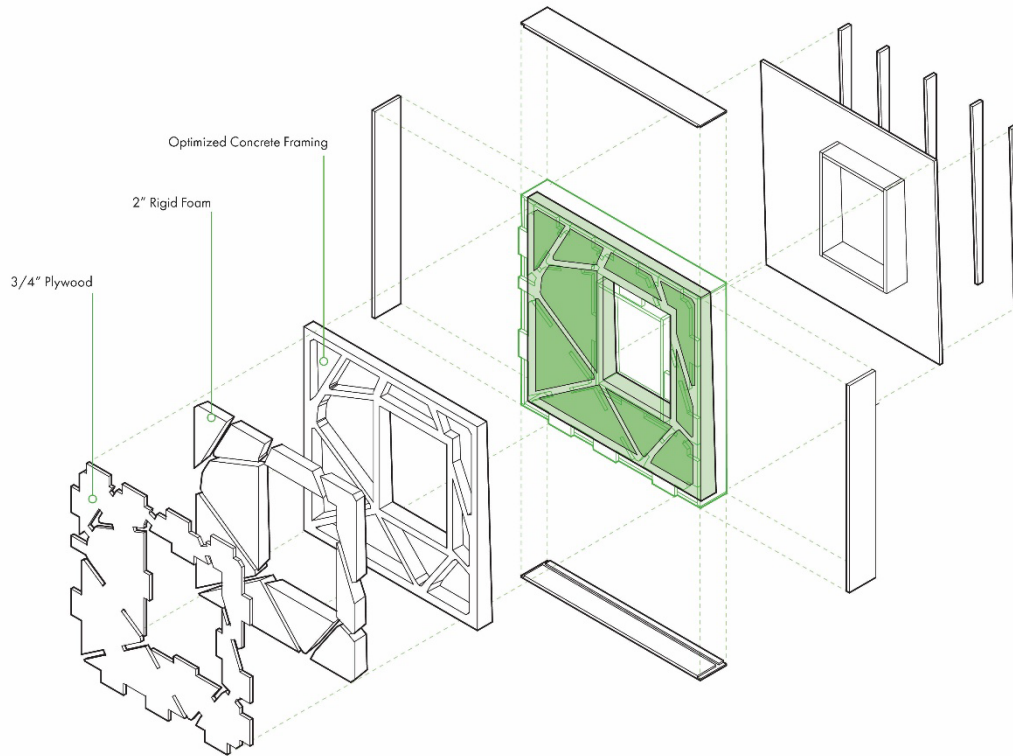


Figure 5. Optimized Wall Panel Digital Assembly. Designed with digital software, Rhinoceros, the panel production is broken into 2 parts, the form work constructed with $\frac{3}{4}$ " plywood and voids created with rigid foam.



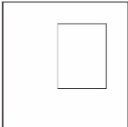
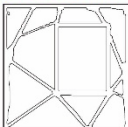
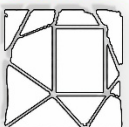
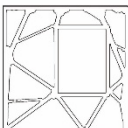
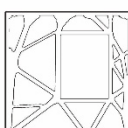
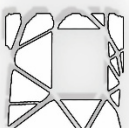
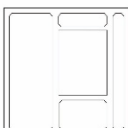
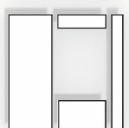
Figure 6. Optimized Wall Panel Fabrication. Fabricated out of $\frac{3}{4}$ " plywood and laminated rigid foam volumes cut to match the angle of the wall. The casting process deals with part 1, casting the concrete into the foam work, and part 2, removing the foam volumes and filling in foam-crete.



Figure 7. Optimized Wall Panel. A 4'-0" x 4'-0" optimized composite wall panel with a concrete outer shell and branches design only where structurally needed and non-structural filler zones of foam-crete results in a panel weight of approximately 156 kg.



Figure 8. King-Stud Wall Panel. A comparable panel based on traditional king-stud framing was cast to compare structural loading performance. With a concrete outer shell and non-structural filler zones of foam-crete, the panel weighs approximately 168 kg.

% Concrete to foam-crete (2D)				Total Mass (kg)	Embodied Carbon (EC) (kgCO ₂ e)	Transit (EC) (kgCO ₂ e)	Total (EC) (kgCO ₂ e)
100%				233	27.69	75.63	103.32
25%				156	22.32	50.75	73.07
30%				162	22.72	52.59	75.31
35%				168	23.14	54.54	77.68
35%				168	23.14	54.54	77.68
			Wall Composite	Foam-crete	Concrete		

33%
Lighter

30%
Carbon
Savings

(26) 4' X 4' Panels
(1) 150km trip on a
flatbed truck

Figure 9. Embodied Carbon. Embodied carbon implications based on degree of optimization and considering a production batch of 26 precast panels (equivalent to a single flatbed truck load arranged in one layer) transported from the manufacturing facility and recycling source to a building site 150 km away (one-way trip).

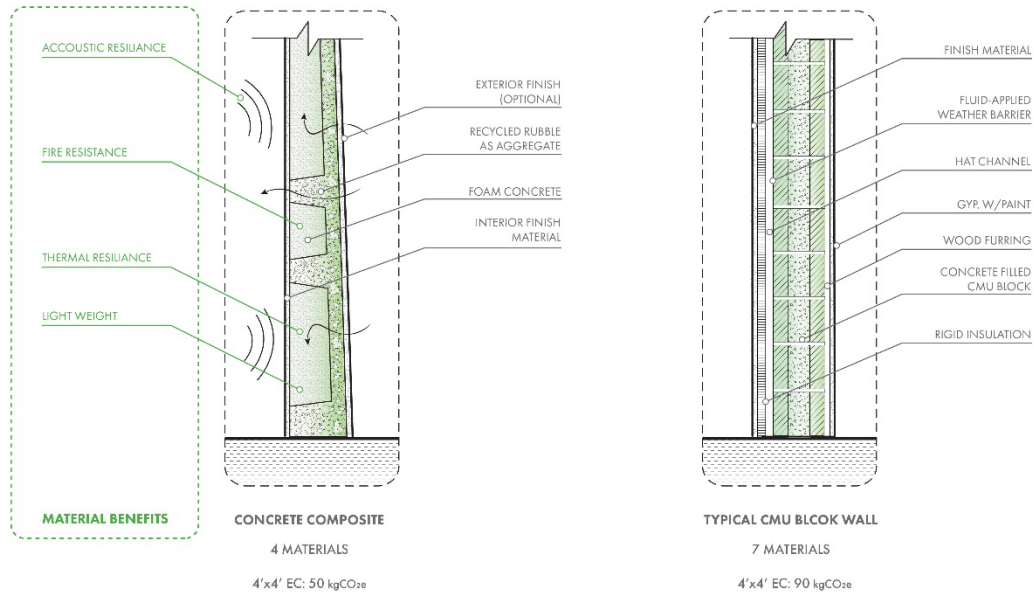


Figure 10. Wall Comparison. The optimized precast panels have potential applications across a range of architectural typologies. The details reveal the difference in the number of materials needed in a typical concrete block wall assembly compared to a concrete composite wall. That said, this study evaluates only embodied emissions and does not account for operational performance (e.g., thermal behavior).